

* errors $|true - approx|$

* static errors $|f(x) - p(x)| \sim \left| \int_a^b f(x) dx - \sum_{i=1}^n p(x_i) \right|$

* sequential errors $|x - x_n| \sim \left| \int_a^b f(x) dx - R_{n,j} \right|$

fixed point approx

Romberg Integration /

root finding

Aitken Extrapolation
Richardson

Jacobian iteration

$\|x - x^{(k)}\|$

sequences of vector

$Ax^{(k)} = b$

$\{x_i\}_{i=0}^m$ where $x_k = x_0 + k \cdot h$ ← step size

If 3 data points step size is half

say my poly interpolation

has 3 data pts has error $O(h^2) + O(h^4)h \dots$

say 5 pt interpolation has error $- O(h^4) + \dots$

result is $O(h^4)$

* fixed point & root finding

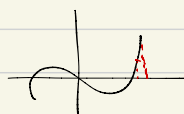
$\exists, !$ (existence, uniqueness)

quadratic convergence

convergence

(bracketing)

converging on a fp of a func



bracketing refers to

which side we're iterating from

$[x^l, x^r]$

moving only 1 side of bracket

(left or right)

Interpolation poly:

coeffs & basis

$p(x) = a_n x^n + \dots + a_1 x + a_0$

Lagrange $p(x) = \sum_{i=0}^n f(x_i) f_i(x)$ require $p(x_i) = f(x_i)$

Hermite poly require $p(x_i) = f(x_i)$ & $p'(x_i) = f'(x_i)$

Cubic poly requires $p(x_i) = f(x_i)$ $s_1^{(k)}(x_{i,0}) = s_{i,1}^{(k)}(x_{i,1})$

Vandermonde $m \times n$ (CVA)

divided diff (diff & integrate on this) (calc connec) *

Do of oscillation [when is an interpolating polynomial exact?]

* end of 1

what are key & what's been tested

Num Diff

* for Lagrange interp

$$f'(x) \approx p'(x)$$

$$f'(x) - p'(x) = \psi_n'(x) + [x_0, \dots, x_n] + \psi_n(x) + [x_0, \dots, x_n]$$

$\{0, 1\}$

interpolate at 0

$$f'(0) = \frac{f(1) - f(0)}{2} \quad \text{what is error when interpolating at 0}$$

$$f'(\frac{1}{2}) = \frac{f(1) - f(0)}{2} \quad \text{what is error}$$

there should be a cliff in interpolating left end & right end

* left vs middle

* Num Integration

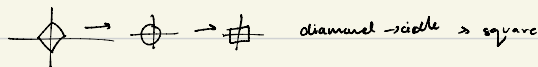
IMV Error

* portion 2

* Numerical Linear Algebra

vector & matrix forms

visualizations (imp for main vector form used in class 11.11.11)



looking at convergence

$$\|x - x^{(k)}\|_{\infty} \text{ for seq } \{x^{(k)}\}$$

- decomposition $A = LU$, $A = LDL^T$, $A = LL^T$ ↙ Cholesky
* conditioned on A

- * LU solve $Ax = b \xrightarrow{\text{transform to}} Ux = \tilde{b}$ then use back substitution

nice way of reducing A via seq

$$A = A^{(1)} \xrightarrow{GE} A^{(n)} = U$$

$$= \begin{bmatrix} u_{11} & \dots & u_{1n} \\ 0 & \ddots & \vdots \\ \vdots & \ddots & u_{nn} \end{bmatrix}$$

$$L = \begin{bmatrix} & \\ & \end{bmatrix}$$

- * DD/SPD/Banded (tri-diagonal)

↓ nice GE properties

↘ create alg solve $Ax = b$ more efficiently

esp pivoting (know what it is why imp)